Databricks Lab : Spark Z-Ordering & Partitioning

Duration : 60 Min.

Scenario

Perform data file compaction and layout co-locality optimization using Partitioning and Z-Ordering.

Input Details

The environment has been pre-configured with the following resources: - **Workspace:** Pre-configured through KodeBuck Service credentials (`DATABRICKS_HOST`, `DATABRICKS_TOKEN`)

Username & Naming Conventions

Your candidate prefix is a combination of your username (the part of your email/login before the `@` or `_` symbol) and exam code. For example, if your username is `student` (or `student_labskraft.com`) and your exam code is `exam123`, your prefix is **`student-exam123`**.

You must name your resources accordingly:

- **Optimized Table:** `<prefix>_zorder_optimize` (e.g. `student_exam123_zorder_optimize`)

Task Objectives

1. Partition and Z-Order

- Compact files and structure partitions.
- Execute Z-ORDER BY query on key columns.

Verification

Once you have performed the tasks, you can run the verification script to check your progress and receive your score. The verification system will check the actual resources in the environment.

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Partition layout check	4 Marks
TC2	Z-order layout metadata check	4 Marks

TC3	Query run metrics check	4 Marks
TC4	Compaction optimization index	4 Marks
TC5	Pruning evaluation check	4 Marks

Total Score: 20 Marks

Important Notes

- Ensure all resource names contain your candidate username prefix in lowercase.
- Check that your script compiles cleanly and contains no syntax errors.